

Header and Footer

Summary

The header/footer microservice is owned by a separate team and bundles search, sign-in, cart, analytics, and consent management. No authoring of header/footer content is required in EDS — only injection.

1. Current State Analysis

The current AEM production architecture uses Apache SSI (**Server-Side Includes**).

Microservice Endpoints

Endpoint	URL Pattern	Used By
Public	https://www.thermofisher.com/global-header-footer/header/nojquery	Apache SSI (production)
Public	https://www.thermofisher.com/global-header-footer/footer/nojquery	Apache SSI (production)
Internal	{gateway}/tf/header/{userType}/{lang}/{country}/noTrustArc.nojquery.shtml	Author/Preview only

The public endpoint returns a **single generic HTML blob** (~500KB) and consists of .

Component	Size
Inline CSS	~200KB
HTML	~50KB
Inline JS	~150KB
Country data	~100KB

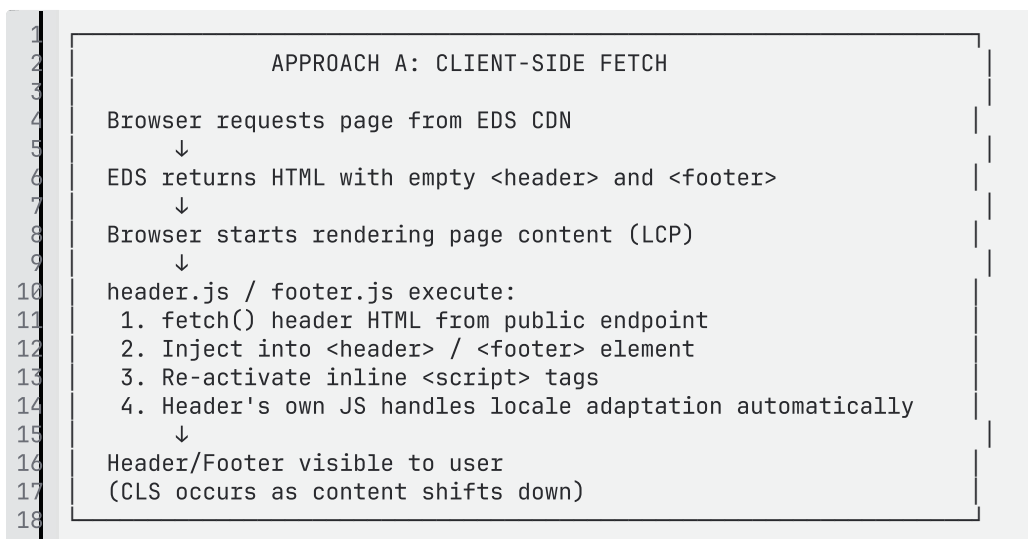
Adobe Launch	External
TrustArc	~50KB

2. EDS Integration

In EDS there are 2 approaches to Implement header

- a. Approach A: Client-Side Fetch
- b. Approach B: Edge Worker Injection

Approach A: Client-Side Fetch



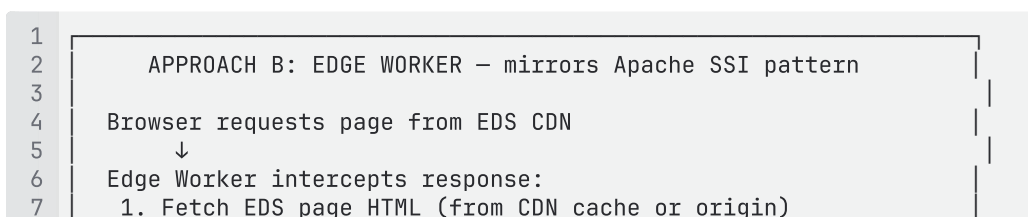
Pros:

- No edge infrastructure required
- Easy to debug (browser DevTools)

Cons:

- **CLS** — page renders, then header pushes content down
- **LCP impact** — +200-500ms before header is visible
- **SEO** — search engines may not see header content
- **Flash of unstyled content** — page loads without header, then it appears

Approach B: Edge Worker Injection



```
8 | 2. Fetch header HTML from public endpoint (from edge cache)
9 | 3. Fetch footer HTML from public endpoint (from edge cache)
10| 4. Stitch: inject header + footer into page HTML
11| 5. Return complete HTML to browser
12|   ↓
13| Browser receives FULL page with header + content + footer
14|
15| No CLS, no extra fetch, SEO-friendly
```

Pros:

- **Zero CLS** — header is in the HTML before browser parses it
- **SEO-friendly** — search engines see complete page
- **Mirrors current SSI pattern** — same cache separation architecture

Cons:

- Requires edge infrastructure setup
- Edge worker latency (~5-20ms) added to TTFB
- Header's ~500KB still impacts page weight

**** Approach B (Edge Worker)** is recommended

3. Assumptions

- All locale adaptation, search, cart, auth, analytics, and consent are handled by the header's own inline JavaScript — automatically, after the page loads in the browser.
- Header looks for cookie `CK_ISO_CODE` , `CK_LANG_CODE` , `identity_uid` to get country, language and usertype and loads header accordingly.
- In EDS, all userType personalization is handled client-side by the header's own JS.

4. Open Questions

- a. Confirm that does Edge worker need to call /global-header and /global-footer url for Microservice or userstate, language,countrycode need to be passed (is Microservice url will always be global path or will have lang, country and usertype, microservice url ?).
- b. Confirm that assumption is correct that for us/en all links will have us/en and for in/en al links will have in/en are handled by header.js .

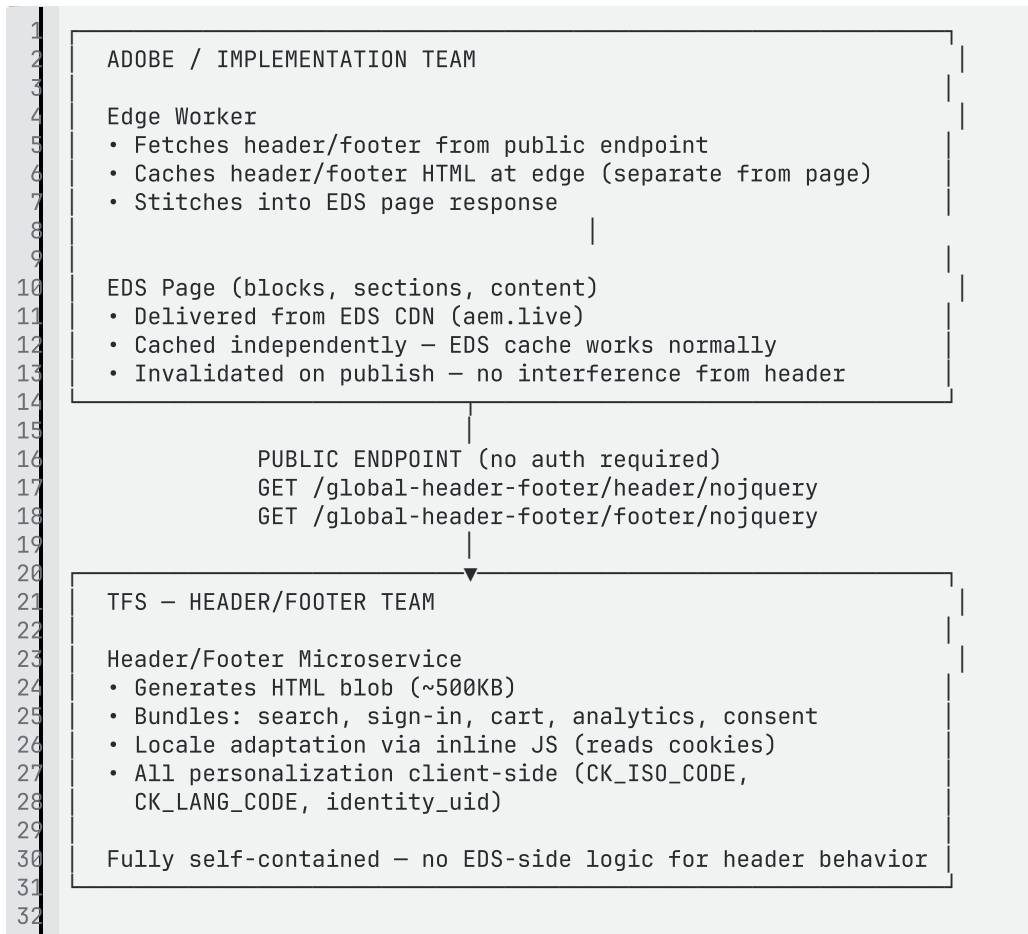
Runtime Ownership & Integration Boundaries

Ownership Matrix

Component	Owner	Responsibility
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Edge Worker (injection logic)	Adobe / Implementation Team	Fetch header/footer from microservice endpoint, stitch into EDS page HTML, manage edge cache for header/footer fragments
Header/Footer Microservice	TFS (Header/Footer Team)	HTML generation, all bundled functionality (search, sign-in, cart, analytics, consent, locale adaptation)
Header/Footer Content	TFS (Header/Footer Team)	Navigation links, search config, language/country data, all visual content

Integration Boundary



Authentication / Security

No authentication is needed for this integration.

Integration Point	Auth?	Reason
Edge Worker → Header/Footer endpoint	None	Public endpoints — same URLs currently used by Apache SSI. No API key, no mTLS.

Cache Strategy

EDS page cache and header/footer cache are independent. They do not interfere with each other.

Cache Layer	What's Cached	TTL	Invalidation
EDS CDN (aem.live)	Page HTML only (blocks, sections, content) — WITHOUT header/footer		Push-based invalidation on publish from AEM
Edge Worker cache	Header HTML (~500KB) + Footer HTML — from microservice	5–15 minutes (or match existing Apache SSI cache)	TTL expiry

Key points:

- All personalization (locale, cart state, sign-in) happens **client-side** via header's inline JS reading cookies (`CK_ISO_CODE` , `CK_LANG_CODE` , `identity_uid`)
- No per-user edge caching needed for header/footer
- EDS page cache works normally — Edge Worker only adds header/footer around the cached page content
- EDS push-based cache invalidation is not affected by header injection

Design principle: The page always renders. Header injection is progressive enhancement via Edge Worker.